Reduced HNN Extensions of von Neumann Algebras

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PLAN OF THE TALK

(1) Objective

(2) HNN Extensions in Group Theory

(3) Reduced HNN Extensions

(4) Construction

(5) Modular Theory

(6) Factoriality

(7) Analysis in Concrete Setups

(8) Ascending HNN Extensions

1. Objective

Given Data:

- v.N.algebra N;
- v. N. subalgebra D;
- injective normal $*$-homo. $\theta : D \to N$;
- $\exists$ f.n.conditional expectations

$$E_D^N : N \to D, \quad E_{\theta(D)}^N : N \to \theta(D).$$

Main objective is to introduce and investigate **the Reduced HNN Extension:**

$$\left(M, E_D^M\right) := \left(N, E_D^N\right) *_D \left(\theta, E_{\theta(D)}^N\right),$$

natural von Neumann algebra analog of the group theoretic one.

2. HNN Extensions in Group Theory

HNN extensions were first appeared in:

H-N-N's Thm. (1949) Each countable discrete group G has a two generator group $\tilde{G}$, into which G can be embedded.

H-N-N = G.Higman, H.Neumann & B.Neumann

The proof of H-N-N:

1st Step:

Enlarge G to a three generator group $\widehat{G}$ by the **amalgamated free product** construction.

2nd Step:

Enlarge $\widehat{G}$ to a two generator group $\tilde{G}$ as an **HNN extension**.

Q.E.D.

Def. (HNN Extensions)

Given data:

- a countable group G;
- a subgroup $H(\subseteq G)$;
- an injective $*$-homo. $\theta : H \to G$.

The universal group G^* generated by G and a new element t with the relation

$$t\theta(h)t^{-1} = h, \quad h \in H,$$

i.e.,

$$G^* = \left\langle G, t : t\theta(h)t^{-1} = h \ (h \in H) \right\rangle,$$

is called the **HNN extension** with **base group** G and **stable letter** t. The group G^* is usually denoted by $G\bigstar_H\theta$.

Two Examples.

1. Baumslag-Solitar Group BS(n, m):

$$\mathrm{BS}(n,m) := \left\langle a, t : ta^n t^{-1} = a^m \right\rangle,$$

i.e., the HNN extension of

- the base group $G = \mathbb{Z} = \langle a \rangle$;
- the subgroup $H = \langle a^m \rangle \cong \mathbb{Z}$;
- the injective $*$-homo. $\theta_m^n(a^m) = a^n$.

Namely,

$$\mathrm{BS}(n,m) = \mathbb{Z} \bigstar_{\langle a^m \rangle} \theta_m^n.$$

2. Richard Thompson's Group F:

$$F := \left\langle x_1, x_2, \dots \ : \ x_i x_j x_i^{-1} = x_{j+1} \ (i < j) \right\rangle.$$

Define

- the "shift" $\varphi : F \to F$ by

$$\varphi(x_i) := x_{i+1}, \quad i = 1, 2, \dots;$$

- $F^2 =$ the subgroup generated by $x_2, x_3, \dots$

So-called *Britton's lemma* (Reduced Word Thm for HNN extensions) enables us to see

$$F = F^2 \bigstar_{F^2} \left(\varphi \big|_{F^2} \right)$$

with the stable letter x_1, and the right-hand is apparently isomorphic to $F \bigstar_F \varphi$.

Such a kind of HNN extensions, i.e., HNN extensions by proper endmorphisms are called

Ascending HNN Extensions.

A Group Theoretic Realization:

(cf. Serre's book [Tree, Springer])

Let

$$\widehat{G} := \cdots \underset{H}{\overset{\text{id}}{\bigstar}}_{\theta} \overset{-1\text{th}}{G} \underset{H}{\overset{\text{id}}{\bigstar}}_{\theta} \overset{0\text{th}}{G} \underset{H}{\overset{\text{id}}{\bigstar}}_{\theta} \overset{1\text{th}}{G} \underset{H}{\overset{\text{id}}{\bigstar}}_{\theta} \cdots$$

$$= \left\langle \underset{\mathbb{Z}}{\bigstar} G \ : \ \iota_n(h) = \iota_{n+1}\left(\theta(h)\right), \ (h \in H) \right\rangle.$$

Here, ι_n = the embedding map of G onto nth component of $\underset{\mathbb{Z}}{\bigstar} G$; and

$$\underset{H}{\overset{\text{id}}{\bigstar}}_{\theta} = \begin{bmatrix} G & & \bigstar & & G \\ & \text{id} \nwarrow & & \nearrow \theta & \\ & & H & & \end{bmatrix}.$$

Let

$\widehat{\theta} :=$ Right Shift on $\widehat{G}$ i.e., $\iota_n(g) \mapsto \iota_{n+1}(g)$,

well-defined because

$$\widehat{\theta}\left(\left\{\iota_n(h)\iota_{n+1}\left(\theta(h)\right)^{-1}\right\}\right) = \left\{\iota_n(h)\iota_{n+1}\left(\theta(h)\right)^{-1}\right\}.$$

The HNN extension $G\bigstar_H\theta$ is realized as the semi-direct product

$$G\bigstar_H\theta = \langle \iota_0(G), t\rangle = \widehat{G} \rtimes_{\widehat{\theta}} \mathbb{Z},$$

where

- ι_n = the embedding map of G onto the nth free component of $\widehat{G}$;
- t = the generator of $\mathbb{Z}\left(\subseteq \widehat{G} \rtimes_{\widehat{\theta}} \mathbb{Z}\right)$.

[Key Relation]

$$t\theta(h)t^{-1} = h, \quad \forall h \in H.$$

$(\because)$

$$\begin{aligned} t \cdot \iota_0(\theta(h)) \cdot t^{-1} &= \widehat{\theta}(\iota_0(\theta(h))) \\ &= \iota_1(\theta(h)) \\ &= \iota_0(h) \end{aligned}$$

since

$$\iota_k(h) = \iota_{k+1}(\theta(h)), \quad h \in H.$$

3. Reduced HNN Extensions

Given Data:

- v.N.algebra N;
- v.N.subalgebra D;
- family Θ of injective normal $*$-homo.s

$$\theta : D \to N \ (\theta \in \Theta);$$

- $\exists$ f.n.condi.exp.s

$$E_D^N : N \to D,$$
$$E_{\theta(D)}^N : N \to \theta(D) \ (\theta \in \Theta).$$

<u>Reduced HNN Extension:</u>

$$\left(M, E_N^M\right) := \left(N, E_D^N\right) \underset{D}{\bigstar} \left(\Theta, \left\{E_{\theta(D)}^N\right\}_{\theta\in\Theta}\right)$$

is a pair of

- v.N.algebra $M = W^* \langle N, u(\theta) \ (\theta \in \Theta)\rangle$;
- f.n.condi.exp. $E_N^M : M \to N$

with the following two conditions (A), (M).

Condition (A) = Algebraic Property:

$$u(\theta)\theta(d)u(\theta)^* = d, \quad \forall d \in D, \ \forall \theta \in \Theta,$$

that is, the unitary $u(\theta)^*$ implements θ, i.e.,

$$\theta = \mathrm{Ad}u(\theta)^*\Big|_D : D \to \theta(D) \subseteq N.$$

$u(\theta)$ is called the **stable unitary** associated with $\theta \in \Theta$.

Reduced Words:

A given word

$$w = u(\theta_0)^{\varepsilon_0} n_1 u(\theta_1)^{\varepsilon_1} n_2 \cdots n_\ell u(\theta_\ell)^{\varepsilon_\ell}$$

in N and the stable unitaries $u(\theta)$, $\theta \in \Theta$, with $\varepsilon_j \in \{\cdot, *\}$ is said to be **reduced form** if, whenever $\theta_{j-1} = \theta_j$, then the following implications hold:

$$\left(\varepsilon_{j-1} = \cdot,\ \varepsilon_j = *\right) \Rightarrow \left(n_j \in \mathrm{Ker} E^N_{\theta_j(D)}\right);$$
$$\left(\varepsilon_{j-1} = *,\ \varepsilon_j = \cdot\right) \Rightarrow \left(n_j \in \mathrm{Ker} E^N_D\right).$$

Example: A word

$$w = u(\theta)^* n_1 u(\theta) n_2 u\left(\theta'\right) n_3 u\left(\theta'\right)^*$$

becomes of reduced form if

$$n_1 \in N^\circ := \mathrm{Ker} E_D^N, \quad n_3 \in N_{\theta'}^\circ := \mathrm{Ker} E_{\theta'(D)}^N,$$

but no need to specify the position of n_2.

Condition (M) = Property of Moments:
For $\forall$ reduced word w in N and the $u(\theta)$'s, one has

$$E_N^M(w) = 0.$$

Operator Algebraic Britton's Lem.

$\left(M, E_N^M\right)$ is characterized by (A) & (M) when $N\,(\subseteq M)$ and the $u(\theta)$'s generate M.

Voiculecu's theory of freeness suggests:

Freeness $\longleftrightarrow$ Reduced Word Thm
for Amalgamated Free Products.

In HNN case,

Condition (M) $\longleftrightarrow$ Reduced Word Thm
for HNN Extensions.

4. Construction

Assume $\Theta = \{\theta\}$ for simplicity !

- $\Delta_2 :=$ the diagonals in $M_2(\mathbf{C})$.

- The inclusion map

$$\iota_1 : D \otimes \Delta_2 \hookrightarrow N \otimes M_2(\mathbf{C});$$
$$\begin{bmatrix} d_1 & \\ & d_2 \end{bmatrix} \mapsto \begin{bmatrix} d_1 & \\ & d_2 \end{bmatrix}.$$

- The f.n.condi.exp.:

$$E_1 : N \otimes M_2(\mathbf{C}) \to D \otimes \Delta_2;$$
$$E_1 := \begin{bmatrix} E_D^N & \\ & E_D^N \end{bmatrix}.$$

- The distinguished embedding map:

$$\iota_\Theta : D \otimes \Delta_2 \hookrightarrow N \otimes M_2(\mathbf{C});$$
$$\begin{bmatrix} d_1 & \\ & d_2 \end{bmatrix} \mapsto \begin{bmatrix} d_1 & \\ & \theta(d_2) \end{bmatrix}.$$

- The f.n.condi.exp.:

$$E_\Theta : N \otimes M_2(\mathbf{C}) \to D \otimes \Delta_2;$$
$$E_\Theta := \begin{bmatrix} E_D^N & \\ & E_{\theta(D)}^N \end{bmatrix}.$$

Denote

$$N^{(2)} := N \otimes M_2(\mathbf{C})$$
$$\Delta_2^D := D \otimes \Delta_2$$

Consider

$$(\mathcal{N}, \mathcal{E}) = \left(N^{(2)}, E_\Theta : \iota_\Theta\right) \underset{\Delta_2^D}{\bigstar} \left(N^{(2)}, E_1 : \iota_1\right).$$

$\lambda_\Theta =$ Embedding of $N^{(2)}$ onto 1st free-part;
$\lambda_1 =$ Embedding $N^{(2)}$ onto 2nd free-part;
$\lambda =$ Embedding of Δ_2^D into $\mathcal{N}$.

Fact. $\lambda_\Theta \circ \iota_\Theta = \lambda = \lambda_1 \circ \iota_1$.

$$\begin{aligned}
\lambda_{\Theta}\left(\begin{bmatrix} d_1 & \\ & \theta(d_2) \end{bmatrix}\right) &= \lambda_{\Theta} \circ \iota_{\Theta}\left(\begin{bmatrix} d_1 & \\ & d_2 \end{bmatrix}\right) \\
&= \lambda\left(\begin{bmatrix} d_1 & \\ & d_2 \end{bmatrix}\right) \\
&= \lambda_1 \circ \iota_1\left(\begin{bmatrix} d_1 & \\ & d_2 \end{bmatrix}\right) \\
&= \lambda_1\left(\begin{bmatrix} d_1 & \\ & d_2 \end{bmatrix}\right).
\end{aligned}$$

- $p := \lambda\left(\begin{bmatrix} 1 & \\ & 0 \end{bmatrix}\right) = \lambda_\Theta\left(\begin{bmatrix} 1 & \\ & 0 \end{bmatrix}\right).$

- $\pi(n) := \lambda_\Theta\left(\begin{bmatrix} n & \\ & 0 \end{bmatrix}\right),\ \forall n \in N$

$$\rightsquigarrow\ \ \pi : N \to p\left(\lambda_\Theta\left(N^{(2)}\right)\right)p \subseteq p\mathcal{N}p.$$

- $u(\theta) := \lambda_1\left(\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}\right)\lambda_\Theta\left(\begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}\right)$

$$\rightsquigarrow \begin{cases} \text{(i) a unitary in } p\mathcal{N}p, \\ \text{(ii) a reduced word of length 2.} \end{cases}$$

- $M = \pi(N) \vee \{u(\theta)\}$.

- $E^M_{\pi(N)}$ denotes the restriction:

$$\left(\text{The Cond.Exp.: } \mathcal{N} \to \lambda_\Theta\left(N^{(2)}\right)\right)\Big|_M.$$

Since

$$p \in \lambda_\Theta\left(N^{(2)}\right), \quad p\lambda_\Theta\left(N^{(2)}\right)p = \pi(N),$$

$E^M_{\pi(N)}$ becomes a condi.exp.: $M \to \pi(N)$.

Verify Condition (A):

$$
\begin{aligned}
&u(\theta)\pi\left(\theta(d)\right)u(\theta)^* \\
&= \lambda_1\left(\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}\right) \times \\
&\qquad \lambda_\ominus\left(\begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}\right)\lambda_\ominus\left(\begin{bmatrix} \theta(d) & \\ & 0 \end{bmatrix}\right)\lambda_\ominus\left(\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}\right) \\
&\qquad\qquad \times \lambda_1\left(\begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}\right) \\
&= \lambda_1\left(\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}\right)\lambda_\ominus\left(\begin{bmatrix} 0 & \\ & \theta(d) \end{bmatrix}\right)\lambda_1\left(\begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}\right) \\
&= \lambda_1\left(\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}\right)\lambda_1\left(\begin{bmatrix} 0 & \\ & d \end{bmatrix}\right)\lambda_1\left(\begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}\right) \\
&= \lambda_1\left(\begin{bmatrix} d & \\ & 0 \end{bmatrix}\right) \\
&= \lambda_\ominus\left(\begin{bmatrix} d & \\ & 0 \end{bmatrix}\right) = \pi(d).
\end{aligned}
$$

Condition (M) follows from the fact:

$u(\theta)$ is a (special) reduced word in $\mathcal{N}$ of length 2

in essence !

Prop.
With $n = \pi(n)$, $\left(M, E_N^M\right)$ satisfies (A) & (M).

Hence, this is a realization of the HNN extension with base algebra N and stable unitary $u(\theta)$ inside a corner algebra of $\mathcal{N}$.

5. Modular Theory

Thm. $\forall$ f.n.s.weight ψ on D,

$$\sigma_t^{\psi\circ E_D^N\circ E_N^M}(u(\theta)) = u(\theta)\left[D\psi\circ\theta^{-1}\circ E_{\theta(D)}^N : D\psi\circ E_D^N\right]_t.$$

Cor. If τ is a trace on N s.t.

$$\tau\circ E_D^N=\tau,\quad \tau\circ E_{\theta(D)}^N=\tau,$$
$$\left(\tau\big|_{\theta(D)}\right)=\left(\tau\big|_D\right)\circ\theta,$$

then $\tau\circ E_N^M$ becomes again a trace.

Continuous Cores:

$$M \overset{E_N^M}{\supseteq} N \overset{E_D^N}{\supseteq} D \rightsquigarrow \widetilde{M} \overset{\widehat{E_N^M}}{\supseteq} \widetilde{N} \overset{\widehat{E_D^N}}{\supseteq} \widetilde{D}.$$

$$\theta : D \to N \rightsquigarrow \tilde{\theta} : \widetilde{D} \to \widetilde{N},$$

$$E^N_{\theta(D)} : N \to \theta(D) \rightsquigarrow \widehat{E^N_{\theta(D)}} : \widetilde{D} \to \theta(\widetilde{D}) = \tilde{\theta}\left(\widetilde{D}\right).$$

Thm.

$$\left(\widetilde{M}, \widehat{E_N^M}\right) \cong \left(\widetilde{N}, \widehat{E_D^N}\right) \underset{\widetilde{D}}{\bigstar} \left(\widetilde{\Theta}, \left\{\widehat{E^N_{\theta(D)}}\right\}_{\theta\in\Theta}\right),$$

where

$$\widetilde{\Theta} := \left\{\tilde{\theta} : \theta \in \Theta\right\}.$$

(∵) Verify (A) & (M).

6. Factoriality

Let

$$\left(M, E_N^M\right) := \left(N, E_D^N\right) \underset{D}{\bigstar} \left(\theta, E_{\theta(D)}^N\right).$$

Prop.

Suppose

- $\exists$ f.n.states $\varphi,\ \varphi_\theta$ on D;
- $\exists$ unitaries $v \in N_{\varphi \circ E_D^N},\ v_\theta \in N_{\varphi_\theta \circ \theta^{-1} \circ E_{\theta(D)}^N}$

s.t.

$$E_D^N\left(v^n\right) = E_{\theta(D)}^N\left(v^n\right) = 0, \quad E_{\theta(D)}^N\left(v_\theta^n\right) = 0$$

as long as $n \neq 0$.

Then,

$$\{v, v_\theta\}' \cap M^\omega \subseteq N^\omega,$$

and in particular,

$$\{v, v_\theta\}' \cap M \subseteq N.$$

Thm.

Suppose

- $\exists$ f.n.states $\varphi,\ \varphi_\theta$ on D;
- $\exists$ unitaries $v \in N_{\varphi \circ E_D^N},\ v_\theta \in N_{\varphi_\theta \circ \theta^{-1} \circ E_{\theta(D)}^N}$

s.t.

$$E_D^N(v^n) = E_{\theta(D)}^N(v^n) = 0, \quad E_{\theta(D)}^N(v_\theta^n) = 0$$

as long as $n \neq 0$.

Then,

$$\mathcal{Z}(M) = \left\{x \in D \cap \theta(D) \cap N' : \theta(x) = x\right\};$$
$$\mathcal{Z}\left(\widetilde{M}\right) = \left\{x \in \widetilde{D} \cap \tilde{\theta}\left(\widetilde{D}\right) \cap \widetilde{N}' : \tilde{\theta}(x) = x\right\};$$
$$M' \cap M^\omega = \left\{x \in D^\omega \cap \theta^\omega(D^\omega) \cap N' : \theta^\omega(x) = x\right\}.$$

7-1. Concrete Setting 1

Let $\theta \notin \mathbb{Q}$.

$$A_\theta := C^*(u, v) \text{ with } uv = e^{2\pi i\theta} vu.$$
$$R := A_\theta'' \text{ w.r.t. the canonical trace } \tau.$$
$$D := \{u\}''(\subseteq R).$$
$$\theta := \text{the } *\text{-iso.: } D \to R \text{ with } \theta(u) = v.$$
$$M := R \underset{D}{\bigstar} \theta \text{ w.r.t. the } \tau\text{-cond.exp.s.}$$

Conclusions:

- M is a type II_1 factor.
- M has no central sequence, i.e., M is a full factor.

7-2. Concrete Setting 2

Given Data:

- v.N.algebras $N_i \supseteq D_i$ $(i = 1, 2, 3)$;
- bijective normal $*$-homo.

$$\theta_{21} : D_1 \to D_2 \ (\theta \in \Theta);$$

- auto. $\theta_3 \in \mathrm{Aut}\,(D_3)$.
- $\exists$ f.n.cond.exp.s $E_{D_i}^{N_i} : N_i \to D_i$ $(i = 1, 2, 3)$.

Assumption: $\forall i = 1, 2$;
$\exists$ f.n.states φ_i, φ_2 on D_i with the property:
$\exists$ unitaries $v_i \in (N_i)_{\varphi_i \circ E_{D_i}^{N_i}}$ s.t.

$$\varphi_i \circ E_{D_i}^{N_i}(v_i^n) = 0, \quad \forall n \neq 0.$$

Set

$$N := N_1 \otimes N_2 \otimes N_3 \supseteq D := D_1 \otimes \mathbf{C}1 \otimes D_3;$$

$$\theta : D \to N \quad \text{defined by}$$
$$\theta\left(d_1 \otimes 1 \otimes d_3\right) := 1 \otimes \theta_{21}\left(d_1\right) \otimes \theta_3\left(d_3\right);$$

$$E_D^N := E_{D_1}^{N_1} \otimes \left(\varphi_2 \circ E_{D_2}^{N_2}\right) \otimes E_{D_3}^{N_3};$$
$$E_{\theta(D)}^N := \left(\varphi_1 \circ E_{D_1}^{N_1}\right) \otimes E_{D_2}^{N_2} \otimes E_{D_3}^{N_3}.$$

Conclusions:

$$\left(M, E_N^M\right) := \left(N, E_D^N\right) \underset{D}{\bigstar} \left(\theta, E_{\theta(D)}^N\right)$$

satisfies

$$\mathcal{Z}(M) = D_3^{\theta_3} \cap \mathcal{Z}\left(N_3\right);$$

$$\mathcal{Z}\left(\widetilde{M}\right) = \left(\mathbf{C1} \otimes \mathbf{C1} \otimes \left(\widetilde{D_3}^{\widetilde{\theta_3}} \cap \mathcal{Z}\left(\widetilde{N_3}\right)\right)\right) \cap \mathcal{Z}\left(\widetilde{N}\right);$$

$$M_\omega \subseteq M' \cap M^\omega = (D_3^\omega)^{\theta_3^\omega} \cap N_3'.$$

Remark: The right-hand sides as well as the T-set $T(M)$ can be computed explicitly in many concrete cases.

7-3. Concrete Setting 3

Let

$$N := \left(\bigotimes_{\mathbb{Z}} \left(Q, \varphi_Q \right) \right) \rtimes_\gamma \mathbb{Z}$$

be the crossed-product by the non-commutative Bernoulli shift γ.

Choose a diffuse abelian subalgebra:

$$D := \bigotimes_{\mathbb{Z}} D_0 \subseteq \bigotimes_{\mathbb{Z}} Q_{\varphi_Q}.$$

Then, $\exists\ \theta : \left(D, \left(\otimes_{\mathbb{Z}} \varphi_Q \right) \big|_D \right) \cong \left(\lambda^\gamma (\mathbb{Z})'', \tau_{\mathbb{Z}} \right)$.

Set

$$E_D^N := E_D^{\otimes_{\mathbb{Z}} Q} \circ E^N_{\otimes_{\mathbb{Z}} Q};$$

$$E^N_{\theta(D)} := \left(\left(\bigotimes_{\mathbb{Z}} \varphi_Q \right) \otimes \mathrm{id} \right) \Big|_N .$$

(The cond.exp.s are chosen w.r.t. $\otimes_{\mathbb{Z}} \varphi_Q$.)

Assumption:

$\exists$ unitary $v \in Q_{\varphi_Q}$ with $\varphi_Q(v) = 0$.

N.B. The classical Bernoulli shift with the equal prob.vec.:

$$\prod_{\mathbb{Z}} \left(\{1, 2, \ldots, n\}, \left(\frac{1}{n}, \frac{1}{n}, \ldots, \frac{1}{n} \right) \right)$$

— O.K.

Conclusion:

$$\left(M, E_N^M \right) := \left(N, E_D^N \right) \underset{D}{\bigstar} \left(\theta, E_{\theta(D)}^N \right).$$

is a full factor of type II_1 or type III_λ with $\lambda \neq 0$.

Remark: If D_0 is a Cartan in Q, then so is D in N. However, $\lambda^\gamma(\mathbb{Z})''$ is known to be a singular MASA in N. Therefore, θ cannot be extended to a global auto. on N !

7-4. Analog of H-N-N's Thm

Prop. Each finite v.N.algebra P has a full type II_1 factor $\widetilde{P}$ generated by two Haar unitaries, into which P can be embedded.

(1) The construction of $\widetilde{P}$ is essentially same as in the original H-N-N's Thm.
(2) Our main results is needed to show that the resulting $\widetilde{P}$ becomes a full type II_1 factor.

8. Ascending HNN Etensions

This part is a work still in progress.

Given Data:

- v.N.algebra N;
- endomorphism $\rho : N \to N$;
- $\exists$ conditional expectation

$$E_\rho : N \to \rho(N).$$

Ascending HNN Extensions:

$$\left(M, E_N^M\right) = (N, \mathrm{id}_N) \underset{N}{\bigstar} (\rho, E_\rho)$$

is called an ascending HNN extension.

Let

$$M \overset{E}{\supseteq} N;$$

$$\langle M, N \rangle := \text{the basic ext.,}$$

$\widehat{E} :=$ the dual op.-valued weight: $\langle M, N \rangle \to M$,

i.e., $\widehat{E}\,(m_1 e_N m_2) := m_1 m_2,\ \forall m_1, m_2 \in M.$

Def. (Relative Følner-type Condition)
$\forall \varepsilon > 0$, $\forall$ finite $\mathcal{F}\,(\subseteq \mathcal{U}(M))$, $\exists$ proj. $f \in \langle M, N \rangle$ s.t.

$$\widehat{E}(f) \in M,$$

$$\|ufu^* - f\|_{L^2(\langle M,N \rangle)} < \varepsilon \cdot \|f\|_{L^2(\langle M,N \rangle)},\ \forall u \in \mathcal{F}.$$

Prop. Let

$$\left(M, E_N^M\right) = (N, \mathrm{id}_N) \underset{N}{\bigstar} (\rho, E_\rho)$$

be an ascending HNN extension.

Suppose $\exists$ f.n.state φ on N s.t.

$$\varphi = \varphi \circ \rho^{-1} \circ E_\rho.$$

Then, $M \supseteq N$ satisfies the relative Følner-type condition.

Remark: The Følner proj. $f \in \langle M, N \rangle$ can be chosen so that $\widehat{E}(f) \in \mathbf{C1}$ in the above prop.

Let $\widehat{\rho}$ be

the non-commutative Bernoulli shift
or
the free Bernoulli shift on

$$\left(\widehat{N},\widehat{\varphi}\right) := \bigotimes_{\mathbb{Z}} (P,\varphi_P) \text{ or } \bigstar_{\mathbb{Z}} (P,\varphi_P)\,.$$

Let ρ be the restriction of $\widehat{\rho}$ to

$N :=$ the "half" of the algebra $\widehat{N}$,

i.e.,

$$(N,\varphi) := \bigotimes_{\mathbb{N}\sqcup\{0\}} (P,\varphi_P) \text{ or } \bigstar_{\mathbb{N}\sqcup\{0\}} (P,\varphi_P)\,.$$

Let

$E_\rho :=$ the $\widehat{\varphi}$-condi.exp. : $\widehat{N} \to N$.

Prop. In the current setup, let

$$\left(M, E_N^M\right) = (N, \mathrm{id}_N) \underset{N}{\bigstar} (\rho, E_\rho)\,.$$

Then,

$$\left(M,\ \varphi \circ E_N^M\right) \cong \left(\widehat{N} \rtimes_{\widehat{\rho}} \mathbb{Z},\ \widehat{\varphi} \circ E_{\widehat{N}}^{\widehat{N} \rtimes_{\widehat{\rho}} \mathbb{Z}}\right).$$

These prop.s explain the relationship between Monod-Popa's and Pestov's counter-examples to Eymard's question on co-amenability in operator algebras.

Monod and Popa's type example obtained by replacing "Bernoulli" shift by free shift:

$$\mathbb{F}_2 = \mathbb{F}(\mathbb{Z}) \rtimes \mathbb{Z} \supseteq \mathbb{F}(\mathbb{Z}) \supseteq \mathbb{F}(\mathbb{Z}_{\geq 0});$$

$$\mathbb{F}(\mathbb{Z}) \rtimes \mathbb{Z} = \mathrm{HNN}\left(\mathbb{F}(\mathbb{Z}_{\geq 0}), \text{restricted free shift}\right).$$

Pestov's one:

$$\begin{aligned}\mathbb{F}_2 =& \mathbb{F}(\mathbb{Z}) \rtimes \mathbb{Z} \\ & \supseteq \mathbb{F}(\mathbb{Z}) = \Gamma_0 \rtimes \mathbb{F}(\mathbb{Z}_{\leq -1}) \\ & \qquad \supseteq \Gamma_0 \left(\supseteq \mathbb{F}\left(\mathbb{Z}_{\geq 0}\right)\right);\end{aligned}$$

$\Gamma_0 :=$ the smallest normal subgroup $\supseteq \mathbb{F}\left(\mathbb{Z}_{\geq 0}\right)$, i.e.,

$$\Gamma_0 = \bigvee_{g \in \mathbb{F}(\mathbb{Z}_{\leq -1})} g\mathbb{F}\left(\mathbb{Z}_{\geq 0}\right) g^{-1}.$$

Addenda: C^*-Setting

(1) Except for the factoriality result, almost all the results (of course, the construction) are still varild even in the C^*-setting.

(2) The embedding and exactness results are derived from the corresponding results (due to Blanchard-Dykema/Dykema-Shlyakhtenko) on amalgamated free products thanks to the construction !